

NORWEGIAN UNIVERSITY OF SCIENCE AND TECHNOLOGY

IT3010 - INDIVIDUAL PEER REVIEW

RESEARCH-BASED INNOVATION METHODOLOGIES IN COMPUTER AND INFORMATION SCIENCE

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# Review of Paper: Gamification Elements of Duolingo and Their Impact on Students' Motivation

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*Group members*

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*Group number:* 13

September 1, 2026 (Spring 2020)

Comments: Comments for teachers of the course

<https://www.ntnu.edu/studies/courses/IT3010>



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## Part 1

| Section      | Evaluation criteria                                                           | Your comment                                                                                                                                                                                     |
|--------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Abstract     | Is the abstract motivating? Is a short motivation for the research given?     | The abstract manages to properly motivate the reader in a compact manner.                                                                                                                        |
|              | Is the practical problem mentioned as part of the motivation?                 | Yes, a decline in motivation is mentioned as the practical problem                                                                                                                               |
|              | Does the abstract say what research strategy/design is used?                  | Yes, it mentions using the survey strategy.                                                                                                                                                      |
|              | Does the abstract say what the findings are?                                  | Yes, it mentions the comparative findings between the four user interaction elements.                                                                                                            |
|              | Does the abstract say what the relevance of the findings is and for whom?     | Yes, it mentions how the data can retain users for Duolingo in the long-term.                                                                                                                    |
| Introduction | Does the introduction have a clear and motivating practical problem?          | Yes, the problem is clear from the research sources verifying it. And improving upon it is a reasonably good motivator.                                                                          |
|              | Does the introduction clearly state the objective of the research?            | Yes.                                                                                                                                                                                             |
|              | Does the introduction have clearly formulated research questions?             | Yes.                                                                                                                                                                                             |
|              | Does the introduction define all central concepts without going into details? | Yes.                                                                                                                                                                                             |
| Background   | Does the paper refer to and properly use 3-4 relevant research papers?        | Yes, more than enough.                                                                                                                                                                           |
|              | Does the paper show clearly how the cited papers are relevant for the RQ?     | Yes, the relevancy for the cited papers are all clearly shown. The citations are not all relevant for the RQ, but they are relevant to support any claim made, and is thereby reasonable to add. |

|             |                                                                                    |                                                                                                                                                    |
|-------------|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods     | Does the paper clearly state what research strategy is chosen?                     | Yes.                                                                                                                                               |
|             | Does the paper clearly state what data generation methods are chosen?              | Yes.                                                                                                                                               |
|             | Does the paper argue for why the strategy/data generation methods are adequate?    | Yes.                                                                                                                                               |
|             | Are the details and the specifics of the methods described adequately?             | Yes.                                                                                                                                               |
|             | Are data analysis methods described and are they adequate?                         | Yes.                                                                                                                                               |
| Results     | Are findings represented in a structured way using themes/graphs/tables?           | Yes, the findings are represented in a graph                                                                                                       |
|             | Are the data convincingly described? Are relevant details in place?                | Yes, the relevant details is described without going in to too much details from the survey.                                                       |
|             | Are the data analyzed in a relevant and proper way related to the RQs?             | Yes.                                                                                                                                               |
| Discussion  | Is the discussion based on the findings and not ungrounded opinions?               | Yes.                                                                                                                                               |
|             | Does the discussion go beyond merely listing the findings? Are findings discussed? | Yes.                                                                                                                                               |
|             | Does the discussion relate to the referenced literature from the background?       | The discussion mentions earlier used references, but no specific ones.                                                                             |
|             | Does the discussion go back to RQs and address them?                               | Yes, it does not mention the research question in its entirety, but it does discuss the key concepts it mentions.                                  |
| Conclusions | Does the paper mention limitations and future work?                                | Yes, several limitations of the research are mentioned.                                                                                            |
| References  | Is the reference list formatted properly and contains all bibliographic data?      | Yes, the references are properly formatted, it fails to mention when it was accessed, which could be useful for future work, but is not essential. |

## Part 2

### Evaluation

I rate the research paper with a strong accept.

### Justification

The research paper focused on finding the effect different parts of the Duolingo program had on its users motivation. The paper managed this by generating data based on users with varying levels of involvement with Duolingo. The generated data showed a clear preference in the user interaction elements, and can thus be used by Duolingo for further improvement. The contribution is therefore good data focusing on the specific designs of the program.

The highest evaluation rating was given as there wasn't any flaws that could justify a lower rating. When assessing the paper it covered all of the necessary fields for each section. The content of each of the sections never trail away from the research question. The methodology and the findings are properly detailed. They base the research on other works, while finding a more niche section they can add more data to.

The structure of the paper is well put together, as it managed to explain in detail all of the necessary criteria put forth by the assessment. The references to the generated data could be improved, as they don't go into the details of the survey itself. The amount of data from each participant, and if the questions had a range of responses or just the two options: positive or negative to the question. There is generally a good variation of sources included in the paper.

Although the discussion mentions the research presented in the background section, it does not reference any specific literature, only that it had been presented in that section. The discussion also does not directly address the research question in the paper. Which was a minor flaw in the assessment. Another point worth mentioning in the discussion is that there is no mention of participant or researcher bias.

Who is the target demographic for the research? The paper specified that the topic of motivation for exchange students would be the focus, however at what age range would the data be applicable, and is the data broad enough to be of interest for different groups as well?

Here a some specific suggestions:

- Address the research question and the background source in the discussion
- Add more details to discussing the survey
- Add more details about target demographic
- Mention if bias plays a part

As a conclusion there are some areas that could be more specifically described as per the assessment criteria, and there is a possibility to expand the discussion section. Otherwise the methodology is well written and thought out and the paper as a whole managed to fulfill its purpose and answer the research question in a well written manner.

### Reviewer's confidence

I would rate my confidence as a 3: medium.